Closed Manifolds with Positive Isotropic Curvature in Dimensions 5,6,7

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1 Introduction

1.1 The positive isotropic curvature condition

Positive isotropic curvature was first introduced by Micallef and Moore [MM88] in their study of the Morse index of harmonic two-spheres and the topology of positively curved manifolds. More precisely, let $R \in \mathcal{C}_B(\mathbb{R}^n)$ be an algebraic curvature tensor on $\mathbb{R}^n$, $n \geq 4$. We say that R has positive isotropic curvature, abbreviated PIC, if

$$R_{1313} + R_{1414} + R_{2323} + R_{2424} - 2R_{1234} > 0$$

for every orthonormal four-frame $\{e_1, e_2, e_3, e_4\}$. If instead

$$R_{1313} + R_{1414} + R_{2323} + R_{2424} - 2R_{1234} \geq 0$$

for every such frame, we say that R is weakly PIC. We denote the cone of weakly PIC algebraic curvature tensors by $\mathcal{C}_{\mathrm{PIC}}$.

Similarly, R is said to be PIC1 if

$$R_{1313} + \lambda^2 R_{1414} + R_{2323} + \lambda^2 R_{2424} - 2\lambda R_{1234} > 0$$

for every orthonormal four-frame and every $\lambda \in [0, 1]$, and PIC2 if

$$R_{1313} + \lambda^2 R_{1414} + \mu^2 R_{2323} + \lambda^2\mu^2 R_{2424} - 2\lambda\mu R_{1234} > 0$$

for every orthonormal four-frame and all $\lambda, \mu \in [0, 1]$. The corresponding weak curvature cones are denoted by $\mathcal{C}_{\mathrm{PIC1}}$ and $\mathcal{C}_{\mathrm{PIC2}}$. These two conditions were introduced and studied by Brendle–Schoen [BS09] and Brendle [Bre08].

Hamilton proved that PIC is preserved by the Ricci flow in dimension four [Ham97]; in arbitrary dimensions, this was proved independently by Brendle–Schoen [BS09] and Nguyen [Ngu10]. Preservation of PIC1 and PIC2 was established by Brendle [Bre08] and Brendle–Schoen [BS09], respectively. A general method is provided by Hamilton's tensor maximum principle [Ham86]: on a closed manifold, a closed convex $O(n)$-invariant curvature condition is preserved by the Ricci flow if it is invariant under the Hamilton ODE

$$\frac{d}{dt}R = Q(R). \tag{1.1}$$

Here

$$\begin{aligned} Q(R)_{ijkl} = &\sum_{p,q=1}^{n} R_{ijpq}R_{klpq} \\ &+ 2\sum_{p,q=1}^{n} \big(R_{ipkq}R_{jplq} - R_{iplq}R_{jpkq}\big). \end{aligned} \tag{1.2}$$

See also [Bre10, Theorem 5.10].

We remark that PIC implies positive scalar curvature, PIC1 implies positive Ricci curvature, and PIC2 implies positive sectional curvature; see [Bre10, Proposition 7.3 and Corollaries 7.15, 7.19].

1.2 The Böhm–Wilking transformation

For symmetric bilinear forms A and B, we use the Kulkarni–Nomizu product

$$(A \wedge B)_{ijkl} = A_{ik}B_{jl} + A_{jl}B_{ik} - A_{il}B_{jk} - A_{jk}B_{il}.$$

We write id for the Euclidean metric, so that $(\mathrm{id} \wedge \mathrm{id})_{ijij} = 2$ for $i \neq j$. The standard curvature decomposition is

$$R = W(R) + \frac{1}{n-2}\operatorname{Ric}_0(R) \wedge \mathrm{id} + \frac{\operatorname{scal}(R)}{2n(n-1)}\,\mathrm{id} \wedge \mathrm{id}, \qquad \operatorname{Ric}_0(R) = \operatorname{Ric}(R) - \frac{\operatorname{scal}(R)}{n}\,\mathrm{id},$$

where $W(R)$ denotes the Weyl curvature tensor. Using this decomposition, Böhm and Wilking [BW08] introduced the linear maps

$$\begin{aligned} l_{a,b}(R) &= R + b\operatorname{Ric}_0(R) \wedge \mathrm{id} + \frac{a}{n}\operatorname{scal}(R)\,\mathrm{id} \wedge \mathrm{id} \\ &= R + b\operatorname{Ric}(R) \wedge \mathrm{id} + \frac{a-b}{n}\operatorname{scal}(R)\,\mathrm{id} \wedge \mathrm{id}\,. \end{aligned} \tag{1.3}$$

They used these maps to construct a family of pinching cones and proved that the normalized Ricci flow on a closed manifold with 2-positive curvature operator converges to a metric of constant positive sectional curvature [BW08, Theorems 1 and 3.1]. We will use this notation throughout the paper.

1.3 Main results

A natural question is how to classify manifolds with PIC, PIC1, or PIC2. We focus on the closed case. In his remarkable paper [Ham82], Hamilton settled the three-dimensional PIC1 case, which is equivalent to positive Ricci curvature. The higher-dimensional PIC1 and PIC2 cases were subsequently settled by Brendle [Bre08, Theorem 2] and Brendle–Schoen [BS09, Theorem 3], respectively: such manifolds are diffeomorphic to spherical space forms. In the PIC case, this paper completes the proof of the following theorem.

Theorem 1.1. *Let M^n, $n \geq 4$, be a closed connected smooth manifold admitting a metric with strictly positive isotropic curvature. Suppose that M contains no nontrivial incompressible $(n-1)$-dimensional spherical space forms. Then M is diffeomorphic to a finite connected sum:*

$$M \cong X_1 \# \cdots \# X_k,$$

where each X_i is either a spherical space form S^n/Γ, or a compact quotient

$$(S^{n-1} \times \mathbb{R})/G$$

by a free, cocompact discrete group of isometries.

First, Hamilton developed a Ricci flow surgery scheme in dimension four [Ham97], which was subsequently completed by Chen–Zhu [CZ06]. Both obtained Theorem 1.1 in dimension $n = 4$ under the assumption that there are no nontrivial incompressible spherical space forms. This restriction was then removed using orbifold surgery in [CTZ12]. Later, Brendle constructed suitable pinching cones and, using surgery theory, proved Theorem 1.1 for $n \geq 12$ under the same assumption [Bre19]. Chen later improved Brendle's construction and proved the cases $n = 9, 10, 11$ [Che24]. Cho further refined the estimates and developed the pinching theory in dimensions $n = 7, 8$, proving the $n = 8$ case, while the gluing step remains unavailable in dimension seven [Cho26]. Huang's higher-dimensional orbifold surgery arguments [Hua23] allow the hypothesis concerning incompressible spherical space forms in Theorem 1.1 to be removed once the required pinching estimate is available.

As a result, Theorem 1.1 is reduced to the construction of the required pinching cones in dimensions $n = 5, 6, 7$, namely Theorem 1.2.

Chow and Wang recently sharpened the Micallef–Moore index estimate for harmonic two-spheres [MM88], proved that in dimensions $n = 5, 6$ the fundamental group is virtually free and obtained the Gromov–Schoen-type classification [CW26].

More recently, Brendle and Tsiamis [BT26] gave a construction of transversally invariant pinching cones valid in every dimension $n \geq 5$.

Theorem 1.2. *Let $5 \le n \le 7$. Then there exist $B = B(n) > 0$ and a family of closed convex cones $C(b), 0 < b < B$, such that each cone $C(b), 0 < b < B$ is transversally invariant under the Hamilton ODE $\frac{d}{dt}R = Q(R)$ and the following properties hold:*

(i) *For any fixed positive real numbers θ and N, there exists a small $b_0 > 0$ such that*

$$(\{R : \mathrm{scal}(R) \ge 0,\ R - \theta\,\mathrm{scal}\,\mathrm{id} \wedge \mathrm{id} \in \mathcal{C}_{\mathrm{PIC}}\} \cap \{R : \mathrm{Ric}_{11} + \mathrm{Ric}_{22} - \theta\,\mathrm{scal} + N \ge 0\}) \subset \{R : R + N\mathrm{id} \wedge \mathrm{id} \in C(b_0)\}.$$

Here the Ricci inequality is required for every orthonormal pair.

(ii) *Let $\{b_j\}$ be an increasing sequence such that $\lim_{j\to+\infty} b_j = B$. Then there exists a sequence of positive real numbers $\{\varepsilon_j\}$ with $\lim_{j\to+\infty} \varepsilon_j = 0$ such that*

$$C(b_j) \subset \{R : R + \varepsilon_j\,\mathrm{scal}\,\mathrm{id} \wedge \mathrm{id} \in \mathcal{C}_{\mathrm{PIC1}}\}.$$

1.4 Outline of the proof

We follow the pinching-cone construction of Brendle [Bre19], Chen [Che24], and Cho [Cho26], joining two families of cones. The first family is essentially the one used by Chen and Cho. In this paper, we modify the second family by adding a scalar-curvature term to its defining inequality. In Section 2, we describe the construction of the two families. In Section 3, we prove that they can be glued together. In Section 4, we prove that both families are invariant under the Hamilton ODE.

2 The construction of the cone

The cone we construct is a combination of two families of cones.

The first family of cones is defined as in Definition 3.1 of [Che24].

For $n = 5, 6, 7$, we define $b_* = 1/20$, and

$$\varepsilon = \begin{cases} 10^{-3}, & n = 5, 6 \\ 10^{-4}, & n = 7. \end{cases}$$

For $0 < b \le b_*$, we associate with b the quantities $\alpha, \gamma, \rho, \omega, A, P_\alpha, Q_\alpha$, defined by

$$\alpha = \alpha(b) = \frac{(2 + (n-2)b)^2}{2(2 + (n-3)b)}\,b, \quad \alpha_* = \alpha(b_*)$$

$$\gamma = \gamma(b) = \frac{b}{2 + (n-3)b},$$

$$\rho = \rho(b) = b - \frac{2(n-1)\gamma(1-2b)}{n^2} - \frac{2(n-1)(1+\gamma)\big[n^2b^2 - 2(n-1)(\alpha - b)(1-2b)\big]}{n^2\big(1 + 2(n-1)\alpha\big)},$$

$$\omega = \omega(b) = (1-\varepsilon)\sqrt{\frac{27(2 + (n-2)b)\,b(1 + (n-2)b)^2}{8n^2\rho^3(2 + (n-3)b)^2}},$$

$$A = A(b) = \frac{2 + 8b}{(n-1)(n-4)} + \frac{4}{n}\big(2b + (n-2)\alpha\big),$$

$$P_\alpha = P_\alpha(b) = \frac{2(1 + (n-2)b)^2}{1 + 2(n-1)\alpha},$$

$$Q_\alpha = Q_\alpha(b) = \frac{2\big[(1 + 2(n-1)\alpha)^2 - (1 + (n-2)b)^2\big]}{n\big(1 + 2(n-1)\alpha\big)}.$$

Definition 2.1. For $0 < b \le b_*$, let $E(b)$ be the set of all algebraic curvature tensors S for which there exists a tensor $T \in \mathrm{Sym}^2(\mathfrak{so}(n))$ such that

(i) $T \ge 0$;

(ii) $S - T \in \mathcal{C}_{\mathrm{PIC}}$ in a generalized sense (see the remark below);

(iii) $\mathrm{Ric}(S)_{11} + \mathrm{Ric}(S)_{22} + \frac{2\gamma}{n}\,\mathrm{scal}(S) \ge 0$ for every orthonormal pair $\{e_1, e_2\}$;

(iv) for every orthonormal frame $\{e_i\}$, we have

$$\mathrm{Ric}(S)_{22} - \mathrm{Ric}(S)_{11} \le \omega^{1/2}\,\mathrm{scal}(S)^{1/2}\left(\sum_{p=3}^{n}(T_{1p1p} + T_{2p2p})\right)^{1/2}.$$

We then define $\tilde{C}(b) = l_{\alpha,b}(E(b))$, where $l_{\alpha,b}$ is the Böhm–Wilking transformation.

Remark 2.2. We use $\mathrm{Sym}^2(\mathfrak{so}(n))$ to denote the space of curvature-type tensors that need not satisfy the first Bianchi identity. We say that $T \in \mathrm{Sym}^2(\mathfrak{so}(n))$ has nonnegative isotropic curvature if

$$T_{1313} + T_{1414} + T_{2323} + T_{2424} + 2T_{1342} + 2T_{1423} \ge 0.$$

Theorem 2.3. *Let $n \in \{5, 6, 7\}$ and $0 < b \le b_*$. Then $\tilde{C}(b)$ is transversally invariant under the Hamilton ODE $\frac{d}{dt}R = Q(R)$.*

Here transversal invariance means that $Q(R)$ lies in the interior of the tangent cone at every nonzero boundary point R. We prove Theorem 2.3 in Section 4.1.

In the second family of cones, the new ingredient is the scalar-curvature term $2h(1-\lambda^2)\,\mathrm{scal}(R)$ in the defining inequality for Z_λ. We add this term to the Ricci correction used in [Cho26].

Definition 2.4. For each $n \in \{5, 6, 7\}$, we set

$$a = b + \frac{n-2}{2}b^2, \quad s = \sqrt{2a}, \quad h = b + 150b^2.$$

For $0 < b \le b_*$ and every orthonormal basis $\{e_i\}$, we define

$$\begin{aligned} Z_\lambda(R) &:= \Phi_\lambda(R) + (1-\lambda^2)H(R) \\ &:= R_{1313} + \lambda^2 R_{1414} + R_{2323} + \lambda^2 R_{2424} - 2\lambda R_{1234} \\ &\quad + (1-\lambda^2)(s(\mathrm{Ric}_{11}(R) + \mathrm{Ric}_{22}(R)) + 2h\,\mathrm{scal}(R)) \end{aligned}$$

We then define

$$\hat{C}(b) = \{R \in \tilde{C}(b_*) : Z_\lambda(l_{a,b}^{-1}R) \ge 0, \forall \lambda \in [0,1].\}$$

Theorem 2.5. *Let $n \in \{5, 6, 7\}$ and $0 < b < b_*$. Then $\hat{C}(b)$ is transversally invariant under the Hamilton ODE $\frac{d}{dt}R = Q(R)$.*

We prove Theorem 2.5 in Section 4.2.

The desired family of cones is

$$C(b) = \begin{cases} \tilde{C}(b), & 0 < b \le b_* \\ \hat{C}(B-b), & b_* \le b < B. \end{cases}, \quad B = 2b_* = \frac{1}{10}.$$

We will prove in the next section that this family of cones is continuous in b.

Once these properties have been established, it follows directly from the definitions that this family of cones satisfies the conditions in Theorem 1.2.

3 Gluing two families of cones

In this section, we prove that $\tilde{C}(b_*) = \hat{C}(b_*)$.

It suffices to prove that, in a fixed orthonormal basis, $l_{\alpha_*,b_*}(R) \in \tilde{C}(b_*)$ implies $Z_\lambda(l_{a,b}^{-1} l_{\alpha_*,b_*}(R)) \geq 0$ at $b = b_*$.

For later use, we record the following identities for $0 < b \leq b_*$. In Proposition 3.2 and its proof, we set $b = b_*$. A direct calculation shows that

$$S := l_{a,b}^{-1} l_{\alpha_*,b_*}(R) = R + \kappa \operatorname{Ric}(R) \wedge \mathrm{id} + \frac{Y}{n} \operatorname{scal}(R)\, \mathrm{id} \wedge \mathrm{id} \tag{3.1}$$

$$\operatorname{Ric}(S) = F \operatorname{Ric}(R) + c \operatorname{scal}(R)\, \mathrm{id} \tag{3.2}$$

$$\operatorname{scal}(S) = \tau \operatorname{scal}(R), \tag{3.3}$$

where

$$\kappa = \frac{b_* - b}{1 + (n-2)b}, \quad Y = d - \kappa = \frac{\alpha_* - a}{1 + 2(n-1)a} - \kappa, \quad d = \frac{\alpha_* - a}{1 + 2(n-1)a}$$
$$F = 1 + (n-2)\kappa, \quad c = \kappa + \frac{2(n-1)}{n} Y, \quad \tau = 1 + 2(n-1)d.$$

Furthermore, we obtain the following expression for $Z_\lambda(S)$:

$$Z_\lambda(S) = \Phi_\lambda(R) + \kappa((1+\lambda^2)\operatorname{Ric}_{11} + 2\operatorname{Ric}_{33} + (1+\lambda^2)\operatorname{Ric}_{22} + 2\lambda^2 \operatorname{Ric}_{44}) + \frac{4Y}{n}(1+\lambda^2)\operatorname{scal} \tag{3.4}$$

$$+ (1-\lambda^2)(s(\operatorname{Ric}_{11} + \operatorname{Ric}_{22}) + 2h \operatorname{scal}) \tag{3.5}$$

$$+ \kappa(1-\lambda^2)(s((n-2)(\operatorname{Ric}_{11} + \operatorname{Ric}_{22}) + 2\operatorname{scal}) + 4(n-1)h \operatorname{scal}) \tag{3.6}$$

$$+ \frac{Y}{n}(1-\lambda^2)\operatorname{scal}(4s(n-1) + 4n(n-1)h), \tag{3.7}$$

where $\operatorname{Ric} = \operatorname{Ric}(R), \operatorname{scal} = \operatorname{scal}(R)$.

We now define

$$I_\lambda(R) := \Phi_\lambda(R) + (1-\lambda^2)K, \quad K = \frac{1}{n-4} \sum_{i=1}^{2} \sum_{p=5}^{n} R_{ipip}.$$

Lemma 3.1. *For $n \geq 5$, we have $I_\lambda \geq 0$.*

Proof. Using the orthonormal frames

$$(e_1, e_2, e_3, \lambda e_4 \pm \sqrt{1-\lambda^2} e_p),$$

we obtain from $R \in \mathcal{C}_{\mathrm{PIC}}$ that

$$R_{1313} + \lambda^2 R_{1414} + R_{2323} + \lambda^2 R_{2424} - 2\lambda R_{1234} + (1-\lambda^2)(R_{1p1p} + R_{2p2p}) \geq 0.$$

Averaging this inequality over p gives the desired result. □

It therefore suffices to prove $Z_\lambda(S) - I_\lambda(R) \geq 0$. Collecting the terms in Equation (3.4) with respect to λ^2 gives

$$Z_\lambda(S) - I_\lambda(R) = (1-\lambda^2)N_0 + \lambda^2 N_1.$$

Here

$$N_0 = (\kappa + sF)(\operatorname{Ric}_{11} + \operatorname{Ric}_{22}) + 2\kappa \operatorname{Ric}_{33} + \left(\frac{4Y}{n} + 2sc + 2h\tau\right)\operatorname{scal} - K$$

$$N_1 = 2\kappa \sum_{i=1}^{4} \operatorname{Ric}_{ii} + \frac{8Y}{n} \operatorname{scal}.$$

Proposition 3.2. *For $n \in \{5, 6, 7\}$, $N_0, N_1 \geq 0$. Therefore $\tilde{C}(b_*) = \hat{C}(b_*)$.*

Before proving the proposition, we need the following lemma to estimate the terms involving Ric and scal. We use the notation

$$\mathfrak{R}(i,j;k,l) = R_{ikik} + R_{jljl} + R_{jkjk} + R_{ilil}.$$

By exchanging indices in the definition of $\mathcal{C}_{\mathrm{PIC}}$, we obtain

$$\mathfrak{R}(i,j;k,l) \geq 0. \tag{3.8}$$

Lemma 3.3. *Let $n \geq 5$. For any $R \in \mathcal{C}_{\mathrm{PIC}}$, we have*

$$\sum_{i=1}^{p} \mathrm{Ric}_{ii} \geq 0,$$

for $3 \leq p \leq n$.

Proof. First, assume that $p = 3$. Define

$$A = \sum_{1 \leq i < j \leq 3} R_{ijij}, \quad B = \sum_{i=1}^{3} \sum_{p \geq 4} R_{ipip}.$$

Averaging Equation (3.8) over $i, j \in \{1,2,3\}, k, l \geq 4$ gives $B \geq 0$.
Averaging Equation (3.8) over $\{i,j,k\} = \{1,2,3\}, p \geq 4$ gives $(n-3)A + B \geq 0$.
Hence

$$\mathrm{Ric}_{11} + \mathrm{Ric}_{22} + \mathrm{Ric}_{33} = 2A + B \geq 0. \tag{3.9}$$

If $p > 3$, Equation (3.9) implies that for all distinct i, j, k we have

$$\mathrm{Ric}_{ii} + \mathrm{Ric}_{jj} + \mathrm{Ric}_{kk} \geq 0.$$

Averaging over all triples in $\{1, 2, \cdots, p\}$ gives the desired nonnegativity. □

Lemma 3.4. *Let $n \geq 5$. For any $R \in \mathcal{C}_{\mathrm{PIC}}$, we have*

$$W := \sum_{i=1}^{4} \mathrm{Ric}_{ii} \geq 0, \quad R_{1313} + R_{2323} \leq \min\{\mathrm{scal}(R)/2, W\}.$$

In particular, $K = (n-4)^{-1} \sum_{p=5}^{n} (R_{1p1p} + R_{2p2p}) \leq \mathrm{scal}(R)/2$.

Proof. (i) First, we prove $R_{1313} + R_{2323} \leq \mathrm{scal}(R)/2$ by induction.

For $n = 5$, observe that

$$\mathrm{scal}(R)/2 - R_{1313} - R_{2323} = \frac{1}{2}(\mathfrak{R}(1,4;2,5) + \mathfrak{R}(2,4;1,5) + \mathfrak{R}(1,3;4,5) + \mathfrak{R}(2,3;4,5)) \geq 0$$

by Equation (3.8).

Suppose the assertion holds for $n = k$. When $n = k + 1$, restricting the curvature tensor to the hyperplane perpendicular to e_p for each $p \geq 4$ and applying the induction hypothesis yields

$$R_{1313} + R_{2323} \leq \mathrm{scal}(R)/2 - \mathrm{Ric}_{pp}\,.$$

Averaging over p gives

$$R_{1313} + R_{2323} \leq \mathrm{scal}(R)/2 - \frac{1}{n-3} \sum_{p=4}^{n} \mathrm{Ric}_{pp} \leq \mathrm{scal}(R)/2.$$

(ii) Next, we prove $R_{1313} + R_{2323} \le W$. Define

$$A = \sum_{1\le i<j\le 4} R_{ijij}, \quad B = \sum_{i=1}^{4}\sum_{p\ge 5} R_{ipip}.$$

Then $W = 2A + B$.

Averaging Equation (3.8) over $\{i, j, k, l\} = \{1, 2, 3, 4\}$ gives $A \ge 0$.

For $n \ge 6$, averaging Equation (3.8) over $i, j \in \{1, 2, 3, 4\}, p, q \ge 5$ gives $B \ge 0$.

Restricting the curvature tensor to a five-frame $\{e_1, e_2, e_3, e_4, e_p\}$ and applying (i) gives

$$R_{1313} + R_{2323} \le A + \sum_{i=1}^{4} R_{ipip}.$$

Averaging over $p \ge 5$ yields

$$R_{1313} + R_{2323} \le A + \frac{1}{n-4} B.$$

Hence $W - R_{1313} - R_{2323} \ge A + \frac{n-5}{n-4} B \ge 0$. The inequality also holds for $n = 5$, since the coefficient of B is zero.

□

Proof of Proposition 3.2. Recall from the definition of $\tilde{C}(b)$ that $R \in E(b_*)$ implies

$$\mathrm{Ric}_{ii}(R) + \mathrm{Ric}_{jj}(R) \ge -\frac{2\gamma(b_*)}{n}\,\mathrm{scal}(R).$$

For N_0, Lemma 3.4 gives

$$N_0 = (\kappa + sF)(\mathrm{Ric}_{11} + \mathrm{Ric}_{22}) + 2\kappa\,\mathrm{Ric}_{33} + \left(\frac{4Y}{n} + 2sc + 2h\tau\right)\mathrm{scal} - K \tag{3.10}$$

$$\ge \left(\frac{4Y}{n} + 2sc + 2h\tau - \frac{1}{2} - \frac{2\gamma(b_*)(sF + 2\kappa)}{n}\right)\mathrm{scal} =: C_0\,\mathrm{scal}; \tag{3.11}$$

For N_1,

$$N_1 = 2\kappa \sum_{i=1}^{4} \mathrm{Ric}_{ii} + \frac{8Y}{n}\,\mathrm{scal} \ge \frac{8Y}{n}\,\mathrm{scal} =: C_1\,\mathrm{scal}\,. \tag{3.12}$$

The constants C_0, C_1 can be calculated directly from $b_* = \frac{1}{20}$.

Explicitly,

$$C_0 = \begin{cases} \frac{42859}{120120} - \frac{\sqrt{43}}{3003}, & n = 5 \\ \frac{14299}{39990} - \frac{2\sqrt{11}}{3999}, & n = 6 \\ \frac{29569}{82544} - \frac{3\sqrt{5}}{5159}, & n = 7 \end{cases}, \quad C_1 = \begin{cases} \frac{43}{30030}, & n = 5 \\ \frac{22}{19995}, & n = 6 \\ \frac{9}{10318}, & n = 7 \end{cases}$$

All these values are positive. □

4 Transversal invariance under the Hamilton ODE

4.1 Proof of Theorem 2.3

We use the first-family coefficients $\alpha, \gamma, \rho, \omega, A, P_\alpha, Q_\alpha$ defined in Section 2, with $0 < b \le b_* = 1/20$. As in [Cho26, Section 4], it suffices to verify [Che24, Lemmas 3.4–3.6] and the two comparisons in Remark 4.7 below; the rest follows the proof of [Che24, Theorem 3.3]. For $n = 7$, the result with $\varepsilon = 1/10000$ follows directly from [Cho26, Theorem 3.3 and Section 4.2], since $1/20 < 131/2000$.

4.1.1 The case $n = 5$

Throughout this subsection, $n = 5$ and $0 < b \le b_*$. Substitution gives

$$\alpha = \frac{b(2+3b)^2}{4(1+b)}, \qquad \rho = \frac{3b(7+22b+18b^2)}{25(1+6b+6b^2)}.$$

Lemma 4.1 (Lemma 3.4 of [Che24] at $n = 5$)**.** *The functions*

$$g_1(b) = \frac{(2+14b+36b^2+27b^3)^2}{2(1+3b)^2(1+6b+6b^2)}, \qquad g_2(b) = \frac{2(1+6b+6b^2)^2}{(1+b)(2+14b+36b^2+27b^3)}$$

are both strictly increasing in b.

Proof. Differentiation and factorization yield

$$g_1'(b) = \frac{(2+14b+36b^2+27b^3)}{\cdot\,(2+48b+315b^2+855b^3+1053b^4+486b^5)} > 0,$$
$$(1+3b)^3(1+6b+6b^2)^2$$

$$g_2'(b) = \frac{2(1+6b+6b^2)(8+44b+99b^2+114b^3+54b^4)}{(1+b)^2(2+14b+36b^2+27b^3)^2} > 0.$$

All coefficients in the numerators and denominators are positive. □

Lemma 4.2 (Lemma 3.5 of [Che24] at $n = 5$)**.** *For $0 < b \le b_*$,*

$$\rho'(b) \ge \rho'(1/20) = \frac{1150521}{1729225} > \frac{4}{9}, \qquad 0 < \rho(b) < b.$$

Proof. Differentiating the closed form of ρ gives

$$\rho'(b) = \frac{3(7+44b+144b^2+216b^3+108b^4)}{25(1+6b+6b^2)^2}, \qquad \rho''(b) = -\frac{24(5+18b+18b^2)}{25(1+6b+6b^2)^3} < 0.$$

Hence ρ' is decreasing, which proves the asserted lower bound. The formula for ρ gives $\rho > 0$, and

$$b - \rho(b) = \frac{4b(1+21b+24b^2)}{25(1+6b+6b^2)} > 0$$

gives $\rho < b$. □

Lemma 4.3 (Lemma 3.6 of [Che24] at $n = 5$)**.** *With $\Lambda = \sqrt{15}A/2$, we have*

$$(1+b\sqrt{3})^2 < 1+\Lambda, \tag{4.1}$$
$$\frac{(1+\Lambda)^2}{P_\alpha + 5Q_\alpha} < \frac{\omega}{4}, \tag{4.2}$$
$$\frac{1+\Lambda}{P_\alpha} < \frac{\omega}{4}. \tag{4.3}$$

Proof. Here

$$A = \frac{5+65b+108b^2+54b^3}{10(1+b)}, \qquad P_\alpha = \frac{2(1+3b)^2}{1+8\alpha}, \qquad P_\alpha + 5Q_\alpha = 2(1+8\alpha).$$

The bounds $\sqrt{3} < 7/4$ and $\sqrt{15} > 15/4$, together with

$$1 + \frac{15}{8}A - \left(1+\frac{7}{4}b\right)^2 = \frac{15+139b+219b^2+113b^3}{16(1+b)} > 0,$$

prove Equation (4.1). Define

$$\mathcal{R}(b) = \frac{999}{2000}\sqrt{\frac{27b(2+3b)}{200\rho^3}}, \qquad \frac{\omega}{4} = \mathcal{R}(b)\frac{1+3b}{2(2+2b)}.$$

The sign of $(\mathcal{R}^2)'$ is the sign of

$$(2+6b)\rho - 3b(2+3b)\rho' < b(2+6b) - \frac{4}{3}b(2+3b) = -\frac{2b}{3}(1-3b) < 0.$$

Thus $\mathcal{R}$ is strictly decreasing.

Since $\sqrt{15}/2 < 1937/1000$, it suffices to compare $\mathcal{R}$ with

$$S_2(b) = \frac{(19685 + 135905b + 209196b^2 + 104598b^3)^2}{50000000(1+3b)^2(1+6b+6b^2)},$$
$$S_3(b) = \frac{(1+6b+6b^2)(19685 + 135905b + 209196b^2 + 104598b^3)}{5000(1+b)(1+3b)^2}.$$

Indeed, direct substitution gives

$$\frac{(1+\frac{1937}{1000}A)^2}{P_\alpha + 5Q_\alpha} = \frac{1+3b}{2(2+2b)}S_2(b),$$
$$\frac{1+\frac{1937}{1000}A}{P_\alpha} = \frac{1+3b}{2(2+2b)}S_3(b).$$

Differentiation yields

$$S_2'(b) = \frac{\begin{array}{c}(19685 + 135905b + 209196b^2 + 104598b^3)\\ \cdot\big(17795 + 176502b + 892341b^2 + 2888208b^3\\ + 4079322b^4 + 1882764b^5\big)\end{array}}{25000000(1+3b)^3(1+6b+6b^2)^2} > 0,$$

$$S_3'(b) = \frac{3\begin{pmatrix}19370 + 224377b + 1024043b^2 + 2496478b^3\\ + 3347136b^4 + 2301156b^5 + 627588b^6\end{pmatrix}}{2500(1+b)^2(1+3b)^3} > 0.$$

At the endpoint,

$$S_2(1/20)^2 < \frac{3523}{50} < \frac{3527}{50} < \mathcal{R}(1/20)^2, \qquad S_3(1/20)^2 < \frac{2619}{100} < \frac{3527}{50}.$$

The opposite monotonicities therefore give $S_2(b) < \mathcal{R}(b)$ and $S_3(b) < \mathcal{R}(b)$. The displayed identities prove Equation (4.2) and Equation (4.3). □

4.1.2 The case $n = 6$

Throughout this subsection, $n = 6$ and $0 < b \le b_*$. Substitution gives

$$\alpha = \frac{2b(1+2b)^2}{2+3b}, \qquad \rho = \frac{b(31+140b+160b^2)}{18(2+15b+20b^2)}.$$

Lemma 4.4 (Lemma 3.4 of [Che24] at $n = 6$)**.** *The functions*

$$g_1(b) = \frac{(2+19b+64b^2+64b^3)^2}{(1+4b)^2(2+15b+20b^2)}, \qquad g_2(b) = \frac{(2+15b+20b^2)^2}{(2+3b)(2+19b+64b^2+64b^3)}$$

are both strictly increasing in b.

Proof. Differentiation and factorization yield

$$g_1'(b) = \frac{\begin{array}{c}(2+19b+64b^2+64b^3)\\ \cdot(14+357b+2892b^2+10208b^3+16640b^4+10240b^5)\end{array}}{(1+4b)^3(2+15b+20b^2)^2} > 0,$$

$$g_2'(b) = \frac{8(2+15b+20b^2)(4+30b+90b^2+133b^3+80b^4)}{(2+3b)^2(2+19b+64b^2+64b^3)^2} > 0.$$

All coefficients in the numerators and denominators are positive. □

Lemma 4.5 (Lemma 3.5 of [Che24] at $n = 6$)**.** *For $0 < b \le b_*$, $\rho'(b) > 4/9$ and $0 < \rho(b) < b$.*

Proof. Differentiation gives

$$\rho'(b) = \frac{31 + 280b + 1220b^2 + 2400b^3 + 1600b^4}{9(2 + 15b + 20b^2)^2}, \qquad \rho'(b) - \frac{4}{9} = \frac{5(3 + 8b)}{9(2 + 15b + 20b^2)^2} > 0.$$

The formula for ρ gives $\rho > 0$, and

$$b - \rho(b) = \frac{5b(1 + 26b + 40b^2)}{18(2 + 15b + 20b^2)} > 0$$

gives $\rho < b$. □

Lemma 4.6 (Lemma 3.6 of [Che24] at $n = 6$)**.** *With $\Lambda = \sqrt{6}A$, we have*

$$(1 + 2b)^2 < 1 + \Lambda, \tag{4.4}$$

$$\frac{(1 + \Lambda)^2}{P_\alpha + 6Q_\alpha} < \frac{\omega}{4}, \tag{4.5}$$

$$\frac{1 + \Lambda}{P_\alpha} < \frac{\omega}{4}. \tag{4.6}$$

Proof. Here

$$A = \frac{6 + 153b + 416b^2 + 320b^3}{15(2 + 3b)}, \qquad P_\alpha = \frac{2(1 + 4b)^2}{1 + 10\alpha}, \qquad P_\alpha + 6Q_\alpha = 2(1 + 10\alpha).$$

Since $12/5 < \sqrt{6}$ and

$$1 + \frac{12}{5}A - (1 + 2b)^2 = \frac{4(6 + 103b + 291b^2 + 245b^3)}{25(2 + 3b)} > 0,$$

we obtain Equation (4.4). Define

$$\mathcal{R}(b) = \frac{999}{2000}\sqrt{\frac{27b(2 + 4b)}{288\rho^3}}, \qquad \frac{\omega}{4} = \mathcal{R}(b)\frac{1 + 4b}{2(2 + 3b)}.$$

The sign of $(\mathcal{R}^2)'$ is the sign of

$$(2 + 8b)\rho - 3b(2 + 4b)\rho' < b(2 + 8b) - \frac{4}{3}b(2 + 4b) = -\frac{2b}{3}(1 - 4b) < 0.$$

Thus $\mathcal{R}$ is strictly decreasing.

Since $\sqrt{6} < 49/20$, it suffices to compare $\mathcal{R}$ with

$$S_2(b) = \frac{(894 + 8397b + 20384b^2 + 15680b^3)^2}{90000(1 + 4b)^2(2 + 15b + 20b^2)},$$
$$S_3(b) = \frac{(2 + 15b + 20b^2)(894 + 8397b + 20384b^2 + 15680b^3)}{300(2 + 3b)(1 + 4b)^2}.$$

Indeed, direct substitution gives

$$\frac{(1 + \frac{49}{20}A)^2}{P_\alpha + 6Q_\alpha} = \frac{1 + 4b}{2(2 + 3b)}S_2(b),$$
$$\frac{1 + \frac{49}{20}A}{P_\alpha} = \frac{1 + 4b}{2(2 + 3b)}S_3(b).$$

Differentiation yields

$$S_2'(b) = \frac{\begin{array}{c}(894 + 8397b + 20384b^2 + 15680b^3)\\ \cdot \big(5874 + 92347b + 641684b^2 + 2372640b^3\\ + 4076800b^4 + 2508800b^5\big)\end{array}}{90000(1+4b)^3(2+15b+20b^2)^2} > 0,$$

$$S_3'(b) = \frac{\begin{array}{c}26436 + 432412b + 2859273b^2 + 9998644b^3\\ + 19208000b^4 + 19004160b^5 + 7526400b^6\end{array}}{300(2+3b)^2(1+4b)^3} > 0.$$

At the endpoint,

$$S_2(1/20) = \frac{2075622481}{403200000} < 6, \qquad S_3(1/20) = \frac{318913}{77400} < 6,$$
$$\mathcal{R}(1/20)^2 = \frac{305002079613}{6553600000} > 36.$$

The opposite monotonicities give $S_2(b) < \mathcal{R}(b)$ and $S_3(b) < \mathcal{R}(b)$. The displayed identities prove Equation (4.5) and Equation (4.6). □

Remark 4.7. The remaining coefficient is

$$\begin{aligned} &nb^2(1-2b) - 2(\alpha - b)(1 - 2b + nb^2)\\ &\quad = \frac{b^2\big(2 + (n-8)b - 2(n+2)(n-2)b^2 - n(n-2)^2b^3\big)}{2 + (n-3)b} > 0. \end{aligned} \tag{4.7}$$

Indeed, for $n = 5, 6$ the derivatives of the polynomial in parentheses are respectively $-3-84b-135b^2 < 0$ and $-2 - 128b - 288b^2 < 0$; its values at $b = 1/20$ are $2783/1600$ and $216/125$. Also, since $A > 0$ and $(1+4b)(n-5)^2(n-2) < (n-1)(n-4)^2$ for $n = 5, 6$,

$$\frac{(1+4b)^2(n-5)^2(n-2)}{(n-1)^2(n-4)^2(1+\Lambda)} < \frac{1+4b}{n-1} < \frac{1}{2}. \tag{4.8}$$

With Lemmas 4.1 to 4.6 and Equations (4.7) to (4.8) verified, the dimension-free remainder of the proof of [Che24, Theorem 3.3] applies unchanged, proving transversal invariance of $\tilde{C}(b)$ for $n = 5, 6$.

4.2 Proof of Theorem 2.5

We prove the transversal invariance of $\hat{C}(b)$ for $0 < b < b_*$, following [Cho26, Che24]. It remains to check $Z_\lambda'(S) > 0$ at every active constraint for $0 \neq l_{a,b}(S) \in \hat{C}(b)$ and $\lambda \in [0, 1]$.

4.2.1 The case $0 \le \lambda \le \dfrac{7}{10}$

The Hamilton ODE gives $\frac{d}{dt}R = Q(R)$, where $R(t) = l_{a,b}(S(t))$. The following lemma gives the evolution equation for $S(t)$.

Lemma 4.8. *Let a, b be fixed, with $1 + (n-2)b \neq 0$ and $1 + 2(n-1)a \neq 0$. If $R(t) \in \mathcal{C}_B(\mathbb{R}^n)$ solves Equation (1.1), then $S(t) = l_{a,b}^{-1}(R(t))$ satisfies*

$$\frac{d}{dt}S = Q(S) + D_{a,b}(S), \tag{4.9}$$

where

$$\begin{aligned} D_{a,b}(S) = {}& (2b + (n-2)b^2 - 2a)\,\mathrm{Ric}_0(S) \wedge \mathrm{Ric}_0(S)\\ &+ 2a\,\mathrm{Ric}(S) \wedge \mathrm{Ric}(S) + 2b^2\,\mathrm{Ric}_0(S)^2 \wedge \mathrm{id}\\ &+ \frac{nb^2(1-2b) - 2(a-b)(1-2b+nb^2)}{n(1+2(n-1)a)} |\,\mathrm{Ric}_0(S)|^2\,\mathrm{id} \wedge \mathrm{id}\,. \end{aligned} \tag{4.10}$$

Moreover,

$$\begin{aligned}\frac{d}{dt}\operatorname{Ric}(S) &= 2(S*\operatorname{Ric}(S)) - 4b\operatorname{Ric}(S)^2 \\ &\quad + \frac{4}{n}(2b+(n-2)a)\operatorname{scal}(S)\operatorname{Ric}(S) \\ &\quad + 2\frac{n^2b^2 - 2(n-1)(a-b)(1-2b)}{n(1+2(n-1)a)}|\operatorname{Ric}_0(S)|^2\operatorname{id} \\ &\quad + \frac{4}{n^2}(a-b)\operatorname{scal}(S)^2\operatorname{id},\end{aligned} \tag{4.11}$$

and

$$\frac{d}{dt}\operatorname{scal}(S) = P_a|\operatorname{Ric}_0(S)|^2 + \frac{2Q_a}{n}\operatorname{scal}(S)^2, \tag{4.12}$$

where

$$P_a = \frac{2(1+(n-2)b)^2}{1+2(n-1)a}, \qquad Q_a = 1+2(n-1)a, \qquad (S*\operatorname{Ric}(S))_{ik} = \sum_{p,q=1}^{n} S_{ipkq}\operatorname{Ric}(S)_{pq}.$$

Proof. These identities are the Böhm–Wilking transformation formulas; see [Bre19, Section 2] and [Cho26, Section 2.1, equations (8)–(10) and Proposition 2.1]. □

We also use the following computational lemma from Proposition A.8 of [Bre19].

Lemma 4.9. *Let $n \geq 5$ and $S \in \mathcal{C}_B(\mathbb{R}^n)$, and let J be a symmetric two-tensor. Suppose that*

$$\Phi_\lambda(S) + (1-\lambda^2)(J_{11}+J_{22}) \geq 0$$

for all orthonormal four-frames and all $\lambda \in [0,1]$, with equality at a given frame and some $\lambda \in [0,1)$. Then, at this frame,

$$\Phi_\lambda(Q(S)) + \Phi_\lambda(J\wedge J) + 2(1-\lambda^2)((S*J)_{11}+(S*J)_{22}) \geq (1+\lambda^2)(J_{11}+J_{22})^2 \geq 0.$$

Define $J = s\operatorname{Ric}(S) + h\operatorname{scal}(S)\operatorname{id}$. Then the hypotheses of Lemma 4.9 are satisfied, and we obtain

$$\Phi_\lambda(Q(S)) \geq (1+\lambda^2)(s(\operatorname{Ric}_{11}+\operatorname{Ric}_{22})+2h\operatorname{scal})^2 - \Phi_\lambda(J\wedge J) - 2(1-\lambda^2)((S*J)_{11}+(S*J)_{22}).$$

Differentiating $Z_\lambda(S)$ gives

$$Z_\lambda'(S) = \Phi_\lambda(S') + s(1-\lambda^2)(\operatorname{Ric}_{11}(S)+\operatorname{Ric}_{22}(S))' + 2h(1-\lambda^2)\operatorname{scal}(S)'.$$

Collecting the terms after setting

$$U = \operatorname{Ric}_{11}+\operatorname{Ric}_{22}, \quad W_0 = (\operatorname{Ric}_0^2)_{11} + (\operatorname{Ric}_0^2)_{22}$$

and using $a = b + \frac{n-2}{2}b^2$ together with

$$\Phi_\lambda(A\wedge\operatorname{id}) = (1+\lambda^2)(A_{11}+A_{22}) + 2A_{33} + 2\lambda^2 A_{44}, \quad \Phi_\lambda(\operatorname{id}\wedge\operatorname{id}) = 4(1+\lambda^2),$$

we obtain

$$Z_\lambda' \geq s^2(1+\lambda^2)U^2 + s(1-\lambda^2)\left(\frac{4(n-2)a}{n}U\operatorname{scal} - 4bW_0 + \frac{8a}{n^2}\operatorname{scal}^2 + 2c_1|\operatorname{Ric}_0|^2\right) \tag{4.13}$$

$$+ 2b^2((1+\lambda^2)W_0 + 2(\operatorname{Ric}_0^2)_{33} + 2\lambda^2(\operatorname{Ric}_0^2)_{44}) + 4c_0(1+\lambda^2)|\operatorname{Ric}_0|^2 \tag{4.14}$$

$$\begin{aligned}&+ 2h\Big\{s((1+\lambda^2)U - 2\operatorname{Ric}_{33} - 2\lambda^2\operatorname{Ric}_{44})\operatorname{scal} \\ &\qquad - (1-\lambda^2)U\operatorname{scal} + (1-\lambda^2)\left(P_a|\operatorname{Ric}_0|^2 + \frac{2Q_a}{n}\operatorname{scal}^2\right)\Big\},\end{aligned} \tag{4.15}$$

where

$$c_0 = \frac{b^2(2-4b-n(n-2)b^2)}{nQ_a}, \quad c_1 = 2b^2 + 2(n-1)c_0.$$

We then normalize by setting $M = \mathrm{Ric}\,/\,\mathrm{scal}$ and define $u = M_{11} + M_{22}, v_i = ((M - \mathrm{id}\,/n)^2)_{ii}$. Collecting terms in Equation (4.13) with respect to u and v_i yields

$$Z'_\lambda \geq \left\{ \begin{array}{l} A_r u^2 + B_r u + C_r + c_V \sum_{j=1}^n v_j + c_W(v_1 + v_2) \\ \qquad + 4b^2 v_3 + 4b^2\lambda^2 v_4 - 4sh(M_{33} + \lambda^2 M_{44}) \end{array} \right\} \mathrm{scal}^2 \tag{4.16}$$

$$=: \mathcal{Q}(\lambda^2)\,\mathrm{scal}^2, \tag{4.17}$$

where

$$\begin{aligned} A_r &= s^2(1+\lambda^2), \\ B_r &= \frac{4(n-2)as}{n}(1-\lambda^2) + h[2s(1+\lambda^2) - 2(1-\lambda^2)], \\ C_r &= \frac{8as}{n^2}(1-\lambda^2) + \frac{4hQ_a}{n}(1-\lambda^2), \\ c_V &= 4c_0(1+\lambda^2) + 2sc_1(1-\lambda^2) + 2hP_a(1-\lambda^2), \\ c_W &= 2b^2(1+\lambda^2) - 4bs(1-\lambda^2), \quad c_U = c_V + c_W. \end{aligned}$$

It is enough to prove that $\mathcal{Q}(\lambda^2) > 0$. Observe that $\mathcal{Q}(\lambda^2)$ is linear in λ^2. Thus it suffices to show $\mathcal{Q}(0) > 0, \mathcal{Q}(49/100) > 0$.

Lemma 4.10. *Assume that $c_U, c_V > 0$ and define $w_0 = c_V + 4b^2, w_1 = c_V + 4b^2\lambda^2$. Then*

$$\mathcal{Q}(t) \geq G(u,t),$$

where $t = \lambda^2$ and

$$\begin{aligned} G(u,t) = A_r u^2 + B_r u + C_r + \frac{c_U}{2}(u - 2/n)^2 - \frac{4sh(1+\lambda^2)}{n} - 4s^2h^2(1/w_0 + \lambda^4/w_1) \\ + \frac{[2/n - u - 2sh(1/w_0 + \lambda^2/w_1)]^2}{1/w_0 + 1/w_1 + (n-4)/c_V}, \end{aligned}$$

is a quadratic function.

Assuming this lemma, the verification is straightforward.
Write

$$\begin{aligned} G(u,0) &= A_2u^2 + A_1u + A_0, \quad \Delta_0 = A_1^2 - 4A_0A_2; \\ G(u,49/100) &= B_2u^2 + B_1u + B_0, \quad \Delta_1 = B_1^2 - 4B_0B_2; \end{aligned}$$

We have

$$c_U > 0, \quad c_V > 0, \quad A_2 > 0, \quad B_2 > 0, \quad \Delta_i < 0, i = 0, 1,$$

as verified by Mathematica.

Proof of Lemma 4.10. Let $N = M - \mathrm{id}\,/n$, and

$$(a_1, a_2, \cdots, a_n) = (c_U, c_U, w_0, w_1, c_V, \cdots, c_V).$$

Then

$$\sum_{i=1}^n a_i v_i = \sum_{i=1}^n a_i \sum_{j=1}^n N_{ij}^2 \geq \sum_{i=1}^n a_i N_{ii}^2 = \sum_{i=1}^n a_i (M_{ii} - 1/n)^2.$$

Using $\sum_{i=1}^n M_{ii} = 1$ and the Cauchy–Schwarz inequality, we obtain

$$\begin{aligned} \sum_{i=1}^n a_i v_i - 4sh(M_{33} + \lambda^2 M_{44}) &\geq \sum_{i=1}^n a_i (M_{ii} - 1/n)^2 - 4sh(M_{33} + \lambda^2 M_{44}) \\ &\geq \frac{c_U}{2}(u - 2/n)^2 - \frac{4sh(1+\lambda^2)}{n} - 4s^2h^2(1/w_0 + \lambda^4/w_1) \\ &\quad + \frac{[2/n - u - 2sh(1/w_0 + \lambda^2/w_1)]^2}{1/w_0 + 1/w_1 + (n-4)/c_V}. \end{aligned}$$

□

4.2.2 The case $\frac{7}{10} \le \lambda \le 1$

For $7/10 \le \lambda \le 1$, we claim that the boundary condition cannot be attained at a nonzero tensor.

Theorem 4.11. *Let $0 \neq l_{a,b}(S) \in \hat{C}(b)$ for some $b \in (0, b_*)$. If $\frac{7}{10} \le \lambda \le 1$, then $Z_\lambda(S) > 0$ in any orthonormal basis.*

Proof. Write $S = l_{a,b}^{-1} l_{\alpha_*,b_*}(R)$ with $R \in E(b_*) \setminus \{0\}$, so that $\operatorname{scal}(R) > 0$. For $\lambda = 1$, Equation (3.4) and Lemma 3.4 give

$$Z_1(S) = \Phi_1(R) + 2\kappa W + \frac{8Y}{n}\operatorname{scal}(R) > 0,$$

since $\kappa \ge 0$, $W \ge 0$, and $Y > 0$ on $0 < b \le b_*$. Now suppose, to the contrary, that in some orthonormal basis $Z_\lambda(S) = 0$ for some $\lambda \in [7/10, 1)$. Then λ is an interior minimizer of $\mu \mapsto Z_\mu(S)$ on $[0,1]$, and differentiation gives

$$\begin{aligned}\frac{1}{2}\frac{d}{d\mu}\Big|_{\mu=\lambda} Z_\mu(S) &= \lambda(S_{1414} + S_{2424} - H) - S_{1234} = 0\\ Z_\lambda(S) &= \lambda^2(S_{1414} + S_{2424} - H) + H + S_{1313} + S_{2323} - 2\lambda S_{1234} = 0.\end{aligned}$$

Recall that $H = H(S) = s(\operatorname{Ric}_{11}(S) + \operatorname{Ric}_{22}(S)) + 2h\operatorname{scal}(S)$.

Eliminating S_{1234}, we obtain

$$\lambda^2(S_{1414} + S_{2424} - H) = S_{1313} + S_{2323} + H.$$

Substituting this identity into $\Phi_1(S)$ gives

$$\begin{aligned}\Phi_1(S) &= S_{1414} + S_{2424} + S_{1313} + S_{2323} - 2S_{1234} &(4.18)\\ &= (1-\lambda)^2(S_{1414} + S_{2424} - H). &(4.19)\end{aligned}$$

To derive a contradiction, it suffices to show $\Phi_1(S) > 9/100(S_{1414} + S_{2424} - H)$.

From the calculation in Equation (3.4), if $S = l_{a,b}^{-1} l_{\alpha_*,b_*}(R)$, we have

$$\Phi_1(S) = \Phi_1(R) + 2\kappa W + \frac{8Y}{n}\operatorname{scal}(R) \ge 2\kappa W + \frac{8Y}{n}\operatorname{scal}(R).$$

Moreover,

$$\begin{aligned}H(S) &= s(F(\operatorname{Ric}_{11}(R) + \operatorname{Ric}_{22}(R)) + 2c\operatorname{scal}(R)) + 2\tau h\operatorname{scal}(R)\\ &\ge \left[2s\left(c - \frac{F\gamma(b_*)}{n}\right) + 2\tau h\right]\operatorname{scal}(R).\end{aligned}$$

By Lemma 3.4 and Equation (3.1),

$$\begin{aligned}S_{1414} + S_{2424} &= (R_{1414} + R_{2424}) + \kappa(\operatorname{Ric}_{11}(R) + \operatorname{Ric}_{22}(R) + 2\operatorname{Ric}_{44}(R)) + \frac{4Y}{n}\operatorname{scal}(R)\\ &\le \left\{\min\{W/\operatorname{scal}(R), 1/2\} + 2\kappa W/\operatorname{scal}(R) + \frac{4\gamma(b_*)\kappa + 4Y}{n}\right\}\operatorname{scal}(R).\end{aligned}$$

Here we use

$$\operatorname{Ric}_{11} + \operatorname{Ric}_{22} + 2\operatorname{Ric}_{44} = 2W - \operatorname{Ric}_{11} - \operatorname{Ric}_{22} - 2\operatorname{Ric}_{33}$$

and the definition of $E(b_*)$.

Combining the inequalities above and setting $T = W/\operatorname{scal}(R)$, we reduce the desired inequality to

$$\frac{1}{n}\left(\frac{191}{25}Y - \frac{9}{50}\gamma(b_*)(2\kappa + sF)\right) + \frac{9}{50}(sc + h\tau) + \frac{91}{50}\kappa T - \frac{9}{100}\min\{T, 1/2\} > 0.$$

(i) If $T \ge 1/2$, it suffices to show that

$$\frac{1}{n}\left(\frac{191}{25}Y - \frac{9}{50}\gamma(b_*)(2\kappa + sF)\right) + \frac{9}{50}(sc + h\tau) + \frac{91}{100}\kappa - \frac{9}{200} > 0,$$

holds for $b \in (0, 1/20)$, as verified by Mathematica.

(ii) If $T \leq 1/2$, we consider the inequality

$$L(T) := \frac{1}{n}\left(\frac{191}{25}Y - \frac{9}{50}\gamma(b_*)(2\kappa + sF)\right) + \frac{9}{50}(sc + h\tau) + \frac{91}{50}\kappa T - \frac{9}{100}T > 0$$

whose left-hand side is linear in T. It suffices to show that

$$L(0) = \frac{1}{n}\left(\frac{191}{25}Y - \frac{9}{50}\gamma(b_*)(2\kappa + sF)\right) + \frac{9}{50}(sc + h\tau) > 0$$
$$L(1/2) = \frac{1}{n}\left(\frac{191}{25}Y - \frac{9}{50}\gamma(b_*)(2\kappa + sF)\right) + \frac{9}{50}(sc + h\tau) + \frac{91}{100}\kappa - \frac{9}{200} > 0$$

hold for $b \in (0, 1/20)$, as verified by Mathematica.

□

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